Scrum 7: Identify Open Source Python Repositories

Objective

The objective of this report is to identify publicly available open-source Python repositories that can serve as representative code samples for research and machine learning model training. The repositories are organized according to their application domains to ensure diversity in software architecture, coding practices, and real-world implementations.

The following repositories have been identified as publicly available open-source Python projects hosted on GitHub.

These repositories contain production-quality Python code and are maintained by active open-source communities.

Acceptance Criterion 1 – List of Open Source Python Repositories

Category	Github Link	Code Review Data Availability
Python Language Implementation	https://github.com/python/cpython	Extensive pull request reviews and maintainer discussions available
Python Programming & Fundamentals	https://github.com/TheAlgorithms/Python	Community pull request reviews and contributor discussions available
Python Standards & Documentation	https://github.com/python/peps	Design discussions, proposal reviews, and maintainer feedback available
Code Formatting	https://github.com/psf/black	Active pull request reviews and code review discussions available
Type Checking	https://github.com/python/mypy	Comprehensive code reviews and contributor feedback available
Testing & Quality Assurance	https://github.com/pytest-dev/pytest	Active pull request reviews and testing-related discussions available

Acceptance Criterion 2 – Repositories Categorized by Application Domain

1. Python Language and Development Ecosystem

2. Web Development

Category	Github Link	Code Review Data Availability
Django	https://github.com/django/django	Extensive pull request reviews, maintainer feedback, and issue discussions available.
Flask	https://github.com/pallets/flask	Active pull request reviews and community code review discussions available.
Pyramid	https://github.com/Pylons/pyramid	Contributor pull request reviews and maintainer feedback available.
Tornado	https://github.com/tornadoweb/tornado	Active pull request reviews, issue discussions, and maintainer code review history available.

3. API Development

Category	Github Link	Code Review Data Availability
FastAPI	https://github.com/fastapi/fastapi / https://github.com/campusx-official/fastapi-course	Extensive pull request reviews, maintainer feedback, and community code review discussions available.
Django REST Framework	https://github.com/encode/django-rest-framework	Active pull request reviews, issue discussions, and contributor feedback available .

Falcon	https://github.com/falconry/falcon	Community pull request reviews, maintainer discussions, and code review history available.
Litestar	https://github.com/litestar-org/litestar	Active pull request reviews, maintainer feedback, and contributor code review discussions available.

4. Machine Learning

Category	Github Link	Code Review Data Availability
ML_Workflow	https://github.com/mlflow/mlflow / https://github.com/krishnaik06/ML-Project	Extensive pull request reviews, maintainer feedback, and community discussions available.
Scikit-learn	https://github.com/scikit-learn/scikit-learn	Active pull request reviews, issue discussions, and comprehensive contributor feedback available.
Feature Engineering	https://github.com/venkatreddy09/Feature-Engineering / https://github.com/feature-engine/feature_engine	Active pull request reviews, issue discussions, and comprehensive contributor feedback available.
Data Preprocessing	https://github.com/Avik-Jain/100-Days-Of-ML-Code/blob/master/Code/Day%201_Data%20PreProcessing.md / https://github.com/scikit-learn/scikit-learn	Data preprocessing modules are actively reviewed through pull requests and maintainer discussions within the Scikit-learn repository.

5. Deep Learning

Category	Github Link	Code Review Data Availability
PyTorch	https://github.com/pytorch/pytorch / https://github.com/mrdbourke/pytorch-deep-learning	Extensive pull request reviews, maintainer feedback, design discussions, and contributor code review history available.
TensorFlow	https://github.com/tensorflow/tensorflow	Active pull request reviews, comprehensive maintainer discussions, and large-scale community code review data available.

6. Web Scraping

Category	Github Repositry	Code Review Data Availability
Scrapy	https://github.com/scrapy/scrapy	Extensive pull request reviews, maintainer discussions, and active contributor code review history available.
Selenium Python	https://github.com/SeleniumHQ/selenium	Large-scale pull request reviews, maintainer feedback, issue discussions, and extensive code review history available.
Requests	https://github.com/psf/requests	Active pull request reviews, contributor feedback, and community code review discussions available.

7. Natural Language Processing (NLP)

Category	Github Repository	Code Review Data Availability

Transformers	https://github.com/huggingface/transformers	Extensive pull request reviews, maintainer feedback, design discussions, and active community code review history available.
NLTK	https://github.com/nltk/nltk	Active pull request reviews, contributor discussions, and maintainer feedback available.
spaCy	https://github.com/explosion/spaCy	Comprehensive pull request reviews, maintainer discussions, and production-quality code review history available.
Sentence Transformers	https://github.com/UKPLab/sentence-transformers	Active pull request reviews, contributor feedback, and community code review discussions available.
Tokenization	https://github.com/huggingface/tokenizers	Extensive pull request reviews, maintainer feedback, design discussions, and active contributor code review history available.

8. Code Parsing Libraries

Category	Github Repository	Code Review Data Availability
Tree-sitter	https://github.com/tree-sitter/tree-sitter	Extensive pull request reviews, maintainer discussions, and contributor code review history available.
Python AST	https://github.com/python/cpython	Extensive pull request reviews, Python core developer feedback, design

		discussions, and code review history available.
LibCST	https://github.com/Instagram/LibCST	Active pull request reviews, maintainer feedback, and community code review discussions available.
astroid	https://github.com/pylint-dev/astroid	Active pull request reviews, maintainer discussions, and contributor code review history available.

9.Model Selection

Project Component	Recommended Model	Github Repository	Reason
Source Code Understanding	**GraphCodeBERT**	https://github.com/microsoft/CodeBERT	Understands program structure, AST, variable relationships, and data flow, making it ideal for analyzing Python code.
Code Review Comment Generation	**CodeT5**	https://github.com/salesforce/CodeT5	Specifically designed for generating natural-language comments, bug explanations, and code summaries.
Defect Detection	**CodeBERT**	https://github.com/microsoft/CodeBERT	Well-suited for identifying code defects and software quality issues.
Intelligent Code Suggestions	**Code Llama**	https://github.com/meta-llama/codellama	Generates contextual coding suggestions and explanations.
Large-Scale Code	**StarCoder2**	https://github.com/	Best for large

Assistant		bigcode-project/starcoder2	codebases and advanced code generation when sufficient GPU resources are available.

Acceptance Criterion 3 – Repository Licenses Permit Research Usage

Repository License Verification

The selected repositories are distributed under widely recognized open-source licenses, including :

- MIT License
- Apache License 2.0
- BSD 3-Clause License
- Python Software Foundation (PSF) License. .

These licenses generally allow researchers to inspect, analyze, modify, and study the source code while complying with the terms specified by each license.

Acceptance Criterion 4 – Repository Metadata Documentation

Repository Metadata

The metadata considered includes:

- Repository Name
- GitHub URL
- Primary Programming Language
- License Type
- Number of Stars
- Number of Forks
- Number of Contributors
- Latest Update Status
- Project Documentation
- Community Activity

These indicators help identify actively maintained repositories that represent real-world software engineering practices.

Acceptance Criterion 5 – Repository Selection Criteria

The repositories included in this report were selected using the following criteria:

- Publicly accessible GitHub repository
- Primary implementation language is Python
- Active open-source project with regular maintenance
- Production-quality Python source code
- License permits research and educational usage
- Well-documented project structure
- Follows established software engineering practices
- Multiple contributors and active community participation
- Represents a distinct application domain
- Suitable for collecting representative Python code samples for software quality analysis and machine learning model training

Conclusion

This report successfully identifies and categorizes representative open-source Python repositories across multiple application domains. The selected repositories are actively maintained, well-documented, and distributed under recognized open-source licenses that support research, software quality analysis, and AI model training.